Speed vs. Security: A Comparison of Standard DNS, DoT, and DoH

Kendra Joachim

December 12, 2025

1 Abstract

While DNS is largely still run in plaintext over UDP by default, alternatives that provide better security are emerging. These protocols, DNS over TLS (DoT) and DNS over HTTPS (DoH), seek to provide better security and privacy, though they are said to be slower and have more overhead due to encryption. This project tested the latency and response byte amount of these protocols and found that DoT and DoH are not significantly slower in most cases and should be implemented whenever possible. For a more secure network, DoT should be implemented, but for more privacy, DoH should be implemented.

2 Introduction

DNS, short for Domain Name Space, is considered the phone book of the Internet. DNS resolvers translate human-readable domain names into machine-readable IP addresses [1]. By default on most browsers, DNS queries and responses are sent in plaintext via UDP. This means that these queries can be read by networks, Internet service providers, and anybody able to monitor transmission. Even if a website uses HTTPS, the query required to navigate to that website is exposed, making it easier for governments to censor the Internet and for attackers to stalk users' online behavior [1]. Two alternatives, DNS over TLS (DoT) and DNS over HTTPS (DoH), have emerged to help mitigate these security issues.

DoT uses the same security protocol (Transport Layer Security or TLS) that HTTPS websites use to encrypt and authenticate communications. It uses TCP with TLS over port 853, which is a dedicated TLS port [2]. With DoH, DNS queries and responses are also encrypted with TLS, but they are sent using TCP over Port 443, which is the port for HTTP traffic [1]. Despite these added measures of security and privacy, Standard DNS is still traditionally used, partially because it is typically the default used on browsers, but also because it claims to have less latency and smaller amounts of response bytes due to its lack of encryption overhead. This project will test those two measurements to

see if they are significant enough to keep Standard DNS as the typical way to send DNS queries and responses.

3 Methods

3.1 Materials

- Python using Visual Studio Code
- Required Libraries:
  - **csv:** used to send results to a comma separated file
  - **datetime:** used to watermark the results
  - **dns.resolver:** used to run standard DNS
  - **httpx:** used to run DNS over HTTPS
  - **matplotlib.pyplot:** used for plotting
  - **subprocess:** used to run DNS over TLS on OpenSSL
  - **time:** used for latency measurements

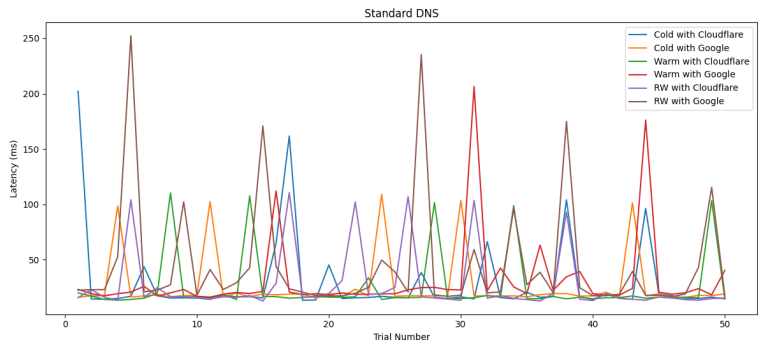
3.2 Process

The general process involved sending DNS queries over the three types of DNS and meausuring the latency and the amount of bytes in the responses. There were limitations in that these tests were only performed on A records because those are the most common records type and are what resolves domain names to IP addresses (IPv4). Further limitations were introduced because Python does not support direct DoT usage through a library. Due to this, it was run over a subprocess using OpenSSL. This meant that only the TCP handshake was able to be simulated in the program, meaning the response byte amount was not able to be tested.

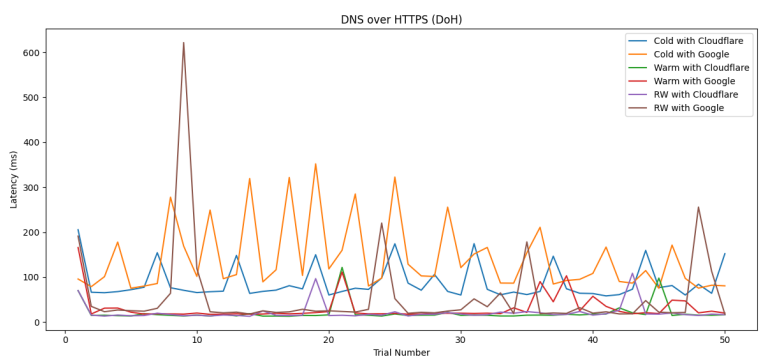
For testing, various conditions were tested to see when each protocol was most effective. The first variation was to run the protocols "cold". This meant that no cache was used and each query was sent as if it was brand new. This was to simulate a worst case scenario. Next, the protocols were run "warm", meaning that there was a cache and the connection remained established and simulating a best case scenario. Both cold and warm used a query to "example.com" for simplicity. Finally, the protocols were run through a "real world" scenario where there was a cache, but the query names were varied. Each url sent for the test was unique with a variety of domains such as .com, .edu, and .uk. This was used as a middle ground between the two cases to simulate what queries would actually be like. It should be noted that due to the nature of DoT only simulating the TCP handshake with TLS, the protocol was only run cold. All three protocols were run under these conditions with Cloudflare (1.1.1.1) and Google (8.8.8.8) as the resolvers. These were also both used as the DoH endpoints.

4 Results

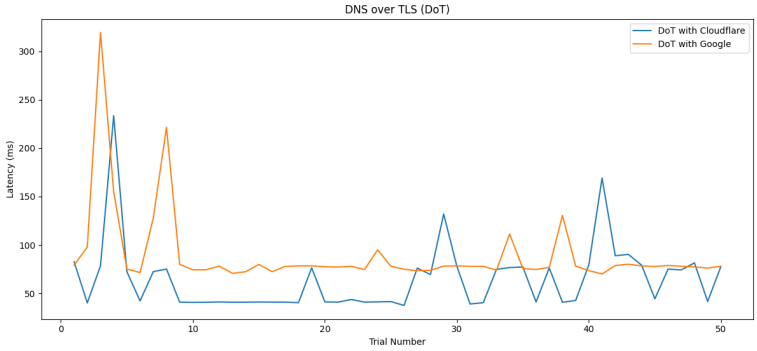
For latency in Standard DNS, real-world with Google, warm with Google, and cold with Cloudflare all had at least one instance of a particularly large latency value, but all other values stayed in the range of 10-100 ms. This does not show that there was much difference in latency between any of the environments or DNS resolvers.



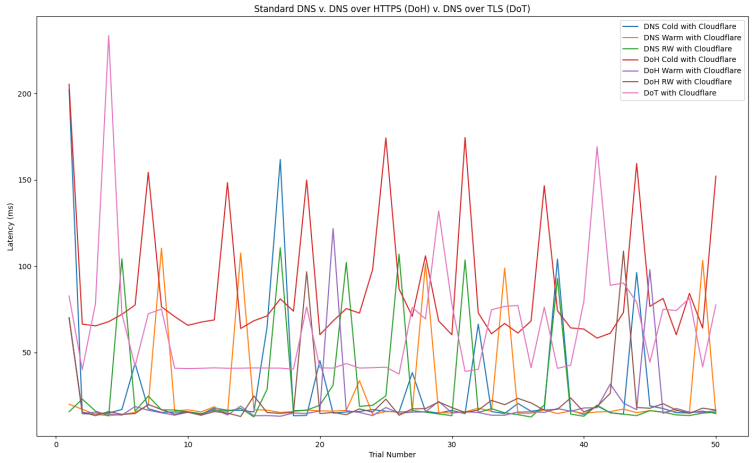
For latency in DoH, it was clear that the cold environment had the biggest latency, ranging from around 50 ms to 300. Besides a few spikes in real-world with Google, the warm and real-world environments appeared to stay stead around 20 milliseconds.



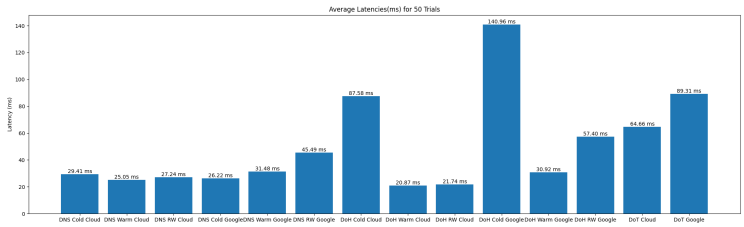
For latency in DoT, the latency tended to stay in a range of 30 to 80 milliseconds with occasional large latency values.



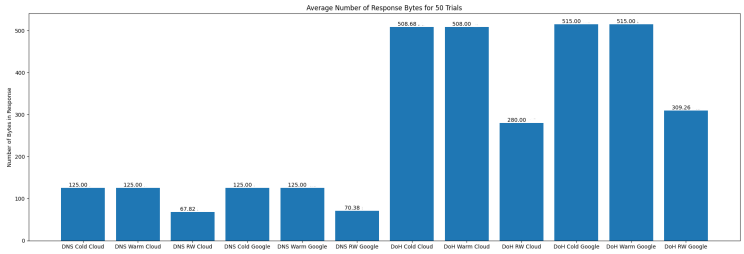
When mapping all of the environments across the three protocols, it can be seen that cold DoH and DoT have consistently larger latency values than the others.



Looking at the average latencies across all environments, protocols, and resolvers, warm DoH with Cloudflare was the fastest while cold DoH with Google was the slowest. When combining resolvers, DNS was fastest for cold and real-world, but DoH was fastest with warm.



In terms of the amount of bytes in a response, warm and cold DoH had the highest number of bytes in responses at a little over 500 bytes in the average response. The real world DNS environments had the least amount of bytes at approximately 70 bytes in a response. Overall, DNS had an average of 106.4 bytes while DoH had an average of 439.3 bytes.



5 Discussion

Overall, Standard DNS over UDP is faster and has a smaller response size. However, DoH was faster in the warm scenario and was only 3 ms slower than Standard DNS in the real world scenario, showing that DoH can perform nearly as fast if not faster than Standard DNS. Given the added security benefits of DoH over Standard DNS, DoH appears to be the better option between the two protocols. Though unable to test DoT on the more likely scenarios (warm and real world), it appears that its performance is likely similar to that of DoH if not better. Thus, due to the added security benefits of DoH and DoT, they should be implemented over Standard DNS when possible.

The key choosing between the two protocols is if network security is the priority or privacy is the priority. Since DoT has a dedicated port, anyone with network visibility can see DoT traffic coming and going even if it is encrypted.

This is bad for privacy as the fact that DNS queries are being sent is still visible [2]. This is a positive for network administrators as this still gives them the ability to monitor and block DNS queries, helping to identify and stop malicious traffic [1]. On the other hand, DoH uses port 443 with all other HTTP traffic. This is great from a privacy perspective because it makes DNS queries indiscernible from other HTTP traffic. This is bad for network administrators because it is infeasible to block any outbound DoH traffic without blocking other HTTP traffic, so it could invalidate protective DNS controls that the company uses [2]. Thus, DoT is better for those concerned with network security while DoH is better for those concerned with privacy.

6 Appendix

GitHub Reposiory: https://github.com/ken-joach/dns-doh-dot

References

[1] https://www.cloudflare.com/learning/dns/dns-over-tls/

[2] https://vercara.digicert.com/resources/what-is-the-difference-between-dnssec-vs-dns-dot-and-doh

[3] https://www.youtube.com/watch?v=HnUDtycXSNE